! CAUTION — NOT ADMITTED BY THE PROTOCOL, EXPLORATORY OOS

Ricochet is **not admitted** by the anti data-mining protocol: its out of sample is exploratory only. Out of sample, from 26 Oct 2022, its t is **1.95** against the classic 1.96 threshold. The out-of-sample period was opened **only once**, with rules and thresholds fixed in advance. Research, not investment advice.

// RESEARCH NOTE · PAPER 1144

RICOCHET

Buying the Nasdaq after a down day

When the Nasdaq-100 closes its regular session below the open, Ricochet buys one E-mini at 17:05 and trades the rebound until 15:00 the next day, with a 50-point stop, a 100-point target and re-entries at the S1 pivot. Out of sample: +$84,365, t 1.95.

AUTHOR
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DATE
4 November 2026

VERSION
1.0

STRATEGY
1144 NQ 5 grezzo OP800

MARKET
NQ · E-mini Nasdaq-100

BARS
5 minutes · Chicago time

DIRECTION
long only

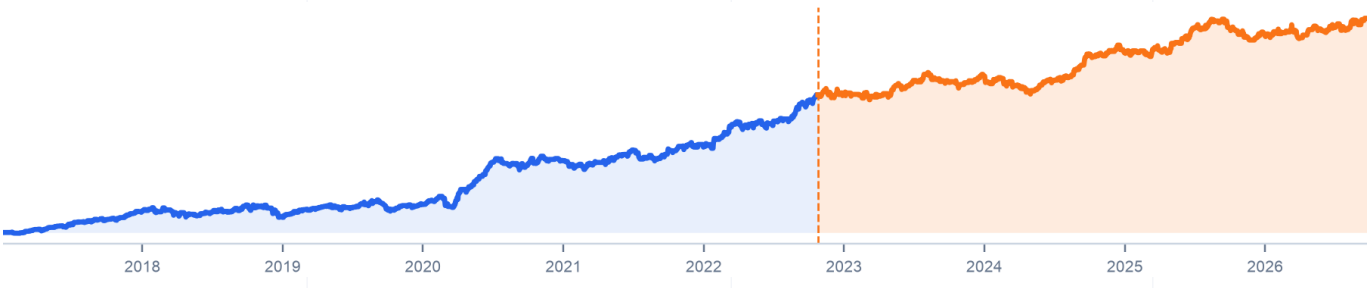
PERIOD
2017 / 2026

CLASSIFICATION
Research note · exploratory OOS

PAPER
Hu, Jin and Wang (2025)

STATUS
not admitted

// EQUITY · 2017-2026 · 1 NQ



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// ABSTRACT

ABSTRACT.

A century of US data says that stock market rebounds are more likely after negative days (Hu, Jin and Wang, 2025). Ricochet turns this into an overnight rule on the E-mini Nasdaq-100: if yesterday's regular session closed below its open, buy at 17:05 Chicago time; stop at 50 points, target at 100, up to three re-entries at the S1 pivot, time exit at 15:00. **Long only, always with a stop.** Development 2017-2022: +$151,843 on 1,168 trades, t 3.73. Out of sample, from 26 Oct 2022 to Sep 2026, opened once: **+$84,365 on 1,070 trades, $78.85 per trade, t 1.95**, one step from the classic 1.96 threshold, with every year positive. The honest part: the raw rebound is barely visible in the data, and most of the profit comes from the re-entries found by the optimisation. A case study worth following forward.

// READ THIS FIRST

Not validated. Positive out of sample (+$84,365) and one step from the classic threshold, with a stop; the paper's premise is barely visible and the profit comes from re-entries. A case study to follow forward.

01

// WHAT IT IS AND WHAT WE FOUND

SUMMARY.

// KEY FIGURES · 1 NQ CONTRACT, COSTS INCLUDED

// OUT-OF-SAMPLE T	// OUT-OF-SAMPLE NET	// MEAN PER TRADE	// DEVELOPMENT T
1.95	**$84,365**	**$78.85**	**3.73**
from 26 Oct 2022 · threshold 1.96	1,070 trades · costs included	out of sample · AV threshold $70	2017-2022 · 1,168 trades

// STOP · TARGET	// MAX DRAWDOWN	// POSITIVE OOS YEARS	// ADMITTED BY THE PROTOCOL
50 · 100 pts	**$23,801**	**4 of 4**	**no**
−$1,009 · +$1,991 net	out of sample · on sessions	2023, 2024, 2025, 2026	exploratory out of sample

- **The idea.** After a down day, the US stock market tends to bounce. Documented on US returns since 1897 and on the main foreign markets.
- **The rule.** If yesterday's regular session of the Nasdaq-100 closed below its open, buy one E-mini NQ at 17:05 Chicago time. Otherwise, stay out.
- **Risk.** A 50-point stop (−$1,009 net) and a 100-point target (+$1,991 net) on every trade. Every loss has the same size.
- **Re-entries.** After a stop or a target, buy again when the price touches the S1 pivot, up to four trades per session; everything closes at 15:00.
- **Out of sample.** +$84,365 on 1,070 trades, t 1.95, every year positive.
- **The catch.** The rebound itself is small and weak out of sample; the edge lives in the re-entries.

// THE VERDICT

Ricochet is the most balanced strategy of this series: a simple idea, a stop on every trade and a positive out-of-sample period of almost four years. It is **not admitted** by our protocol and the premise is weak in the data, so it is a case study to follow forward, not a strategy to put on an account today.

02

// FROM A CENTURY OF DATA TO THE NASDAQ

THE IDEA.

Hu, Jin and Wang (2025) study the autocorrelation of daily US stock returns from 1897 to 2024. It changes over time, but one feature is stable, and appears in the main foreign markets too: **rebounds are more likely after negative days**. A daily strategy built on this asymmetry earns a higher Sharpe ratio than the index.

Ricochet keeps only the asymmetric part and moves it to the E-mini Nasdaq-100 overnight. Two choices are ours: the down day is measured from the open to the close of the regular session (the paper uses close to close), and the position is taken at the evening reopen, 17:05 Chicago time.

// WHAT WE TAKE FROM THE PAPER, AND WHAT WE ADD

ELEMENT	PAPER	RICOCHET
market	US stock market, daily	E-mini Nasdaq-100, 5-minute bars
signal	negative daily return	regular session closed below its open
position	long the next day	long from 17:05 to 15:00 the next day
risk	-	50-point stop, 100-point target
re-entries	-	S1 pivot, at most four trades (optimisation)

// THE RESEARCH QUESTION

Does the rebound after a down day survive on the Nasdaq futures overnight, and is it what makes Ricochet profitable?

03

// ONE SESSION, STEP BY STEP

THE SETUP ON THE PRICE.

- **Signal.** Yesterday's regular session (08:30-15:00 Chicago) closed below its open, by any amount.
- **Entry.** 17:05 Chicago time, at the evening reopen, one E-mini NQ contract ($20 per point).
- **Stop and target.** 50 points below and 100 points above the entry, checked bar by bar; a bar that opens beyond a level exits at the open.
- **Re-entry.** After a stop or a target, buy again when the price touches the S1 pivot and closes back above it; one bar of pause; at most four trades per session.
- **Time exit.** 15:00 Chicago time.
- **Costs.** $9 per trade, included in every number.

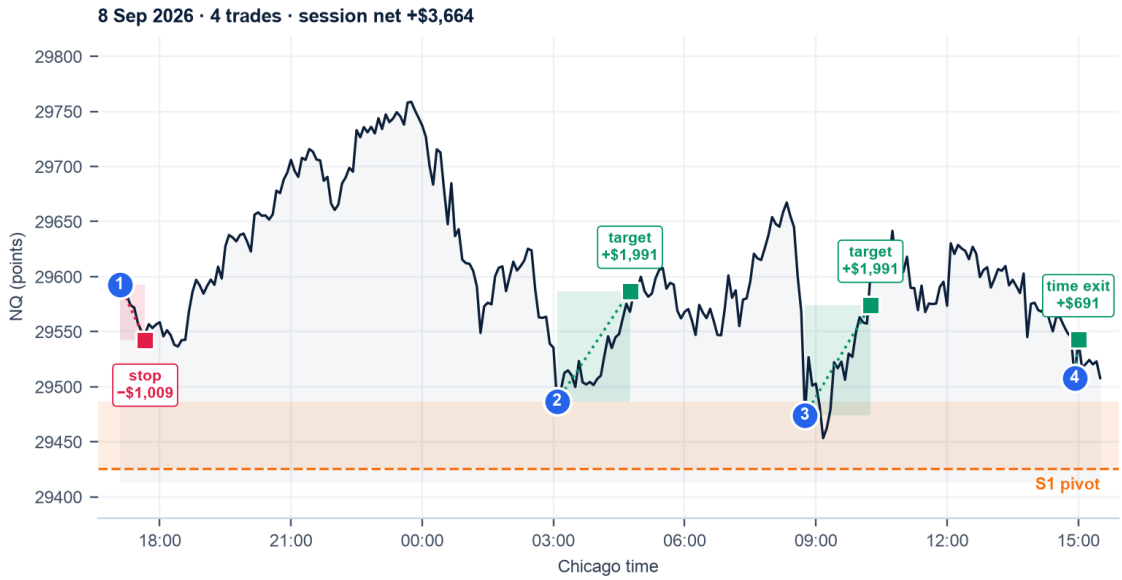


Fig. 1. A session from the out-of-sample period: 8 Sep 2026. Navy: the price; orange: the S1 pivot and, shaded, the re-entry zone just above it; blue dots: entries; squares: exits. Trade 1 enters at 17:05 and is stopped soon after (−$1,009). Overnight the price falls back to the S1 pivot: trade 2 re-enters and reaches the target (+$1,991). After the 08:30 data the price drops again, trade 3 re-enters and hits the target; trade 4 enters near the close and exits at 15:00 (+$691). Net of the session: +$3,664.

// THE DEFINITION, FROZEN BEFORE THE OUT-OF-SAMPLE PERIOD

ELEMENT	VALUE	ORIGIN
signal	regular session closed below its open (threshold 0%)	paper card
entry · time exit	17:05 · 15:00	paper card
direction	long only	paper card
stop · target	$1,000 · $2,000 (50 · 100 points)	paper card
re-entries	S1 pivot, touch and close, at most 4 trades	optimisation
filters, entries, exits, size	none improved the rule	optimisation

// DEVELOPMENT AND OUT OF SAMPLE

04 RESULTS.

// RESULTS (1 NQ CONTRACT, COSTS INCLUDED)

	DEVELOPMENT	OUT OF SAMPLE	TOTAL
dates	2017 - 25 Oct 2022	26 Oct 2022 - Sep 2026	2017-2026
trades	1,168	1,070	2,238
sessions with trades	656	451	1,107
net $	151,843	84,365	236,208
mean per trade $	130.00	78.85	105.54
winning trades	44.4%	37.3%	41.0%
profit factor	1.25	1.13	1.19
max drawdown $	14,581	23,801	23,801
t on sessions	3.73	1.95	3.97

The out-of-sample period was opened once. Our protocol also reads it in two parts (to Dec 2023: t 0.93 against 1.96; from 2024: t 1.71 against 4.26); this note judges it as one period.

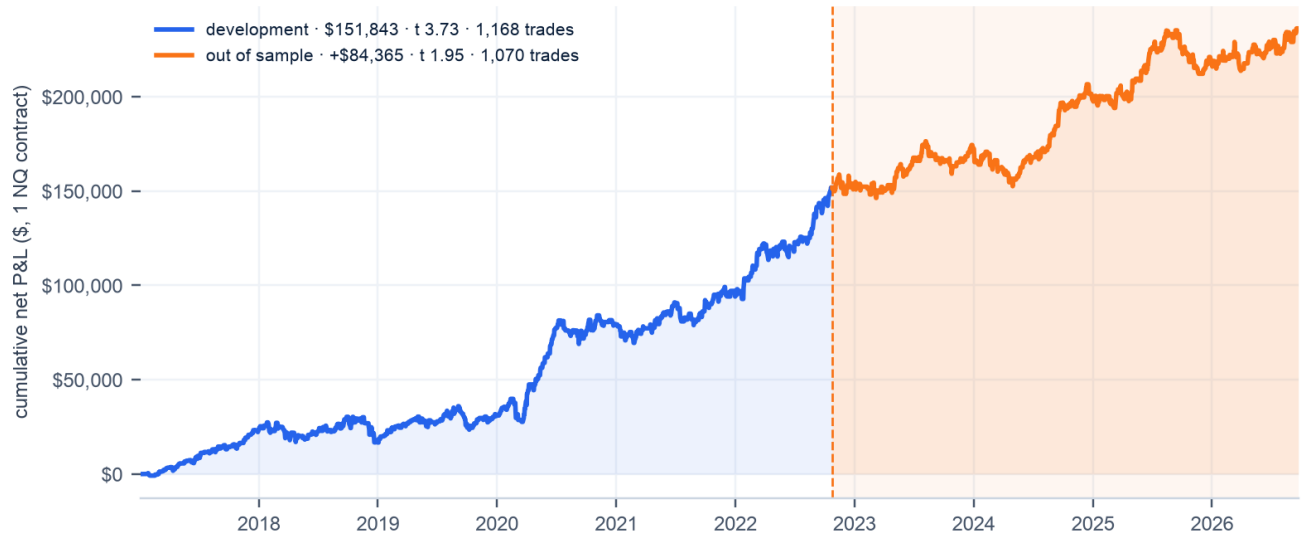


Fig. 2. *Cumulative net P&L; on one E-mini Nasdaq-100 contract, costs included. Blue: development, where the rules were chosen. Orange: out of sample, never seen while choosing. The dashed line is 26 Oct 2022.*

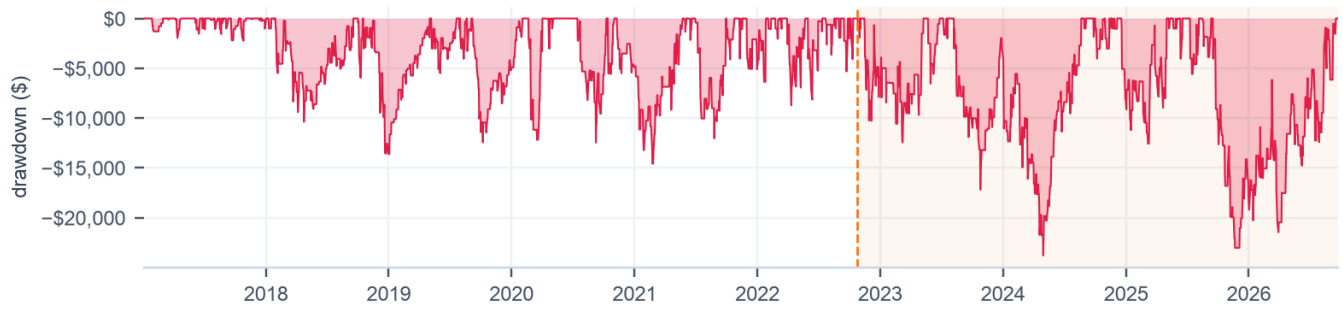


Fig. 3. Drawdown from the previous high. The deepest, $23,801, is in the out-of-sample period, in 2023-2024.

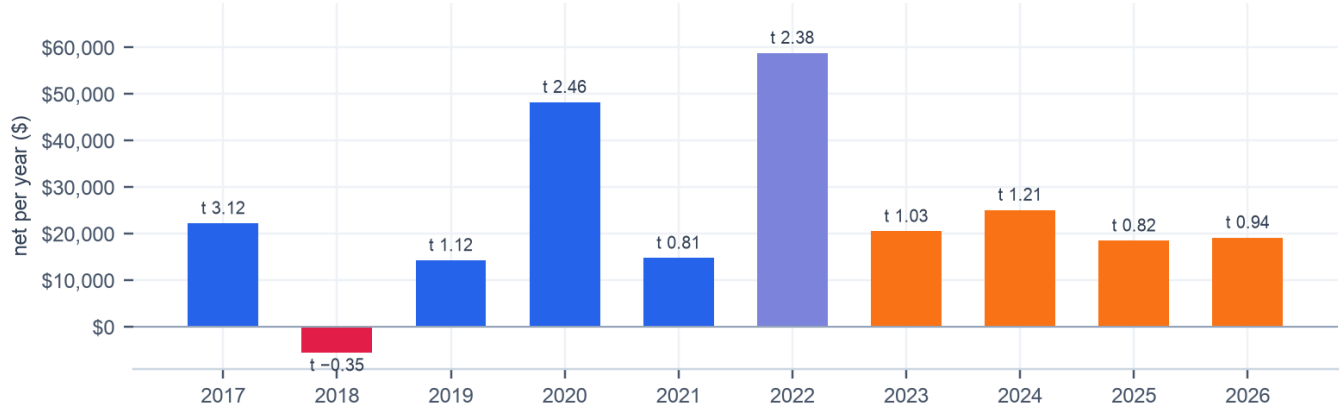


Fig. 4. Net per year with the t of each year. Out of sample every year is positive, between $18,511 and $25,114 a year; no single year has a t above 1.3.

// READING THE CURVE

In development the curve climbs faster than out of sample: the mean per trade drops from $130 to $79 and winning trades from 44% to 37%. The curve keeps rising, more slowly. That is the normal cost of moving from the data used to choose to data never seen.

// THE REBOUND, THE EXITS, THE RE-ENTRIES

05 WHERE THE PROFIT IS.



Fig. 5. The premise, without the strategy: the mean move of NQ from 17:05 to 15:00 after a down day (green) and after an up day (grey). In development the rebound is there (+9.2 points, t 1.35); out of sample it almost disappears (+2.1 points, t 0.15).



Fig. 6. Left: how trades end; light, development; solid, out of sample. Stops outnumber targets about two to one, as the 1:2 risk-reward implies. Right: net by trade number in the session; blue, development; orange, out of sample. The first trade, the one that follows the signal alone, makes little; the profit comes from the re-entries.

// NET BY TRADE NUMBER

TRADE	DEVELOPMENT $	T	OUT OF SAMPLE $	T
1st (17:05)	5,101	0.16	33,681	1.10
2nd	72,202	3.10	53,392	2.19
3rd	39,213	2.33	−1,285	−0.07
4th	35,327	2.96	−1,423	−0.09

// WHAT THIS MEANS

The paper's rebound alone would not pay: the first trade earns $5,101 in six years of development. Ricochet works because, on a day that started weak, buying again at the S1 pivot catches the turn. Out of sample the second trade carries the result; the third and fourth add nothing.

06

// WHAT THE DATA SUPPORT

HYPOTHESES.

H1

PARTLY SUPPORTED

After a down day the Nasdaq-100 rebounds overnight.

// HOW IT IS TESTED

Mean NQ move from 17:05 to 15:00 after down days against up days.

// OUTCOME

Development +9.2 points (t 1.35) against −3.6; out of sample +2.1 (t 0.15) against −1.4. The sign is right, the size is small.

H2

NOT SUPPORTED

The profit comes from the signal itself.

// HOW IT IS TESTED

Net of the first trade of each session against the re-entries.

// OUTCOME

First trade: +$5,101 in development (t 0.16), +$33,681 out of sample (t 1.10). Re-entries: the rest.

H3

NOT SUPPORTED

The rule works on the other US equity indices.

// HOW IT IS TESTED

Step 1: the raw rule on every market at 5 minutes, development.

// OUTCOME

YES only on NQ (t 1.58) and ES (t 0.55); NO on the Dow (t 0.23), the Russell 2000 (0.63) and the Nikkei (0.25). The ES version, optimised separately, fails out of sample.

H4

PARTLY SUPPORTED

The strategy holds out of sample.

// HOW IT IS TESTED

One t on sessions from 26 Oct 2022 to the end of the data; classic threshold 1.96.

// OUTCOME

t 1.95, +$84,365, every year positive: one step below the threshold.

// THE CHAIN, STEP BY STEP

07 HOW IT WAS FOUND.

Paper 1144 entered our automated research in November 2026. Step 1 ran the paper's raw rule on 18 futures markets; it earned a YES only on NQ and ES. On NQ the optimisation searched filters, entries, re-entries, confirmations, exits and size in sequence, keeping a family only if its plateau beat the previous step.

// THE OPTIMISATION CHAIN ON NQ AT 5 MINUTES

STEP	CHOSEN	COMBINATIONS	ZONE T
paper rule (raw)	-	1	1.58
filters	none	506,538	-
entries	none	624,240	-
re-entries	S1 pivot	1,205,280	3.28
confirmations	none	51,840	-
exits	none	91,424	-
size	none	100	-

5,867,006 tests in the chain, all counted in our anti data-mining register.

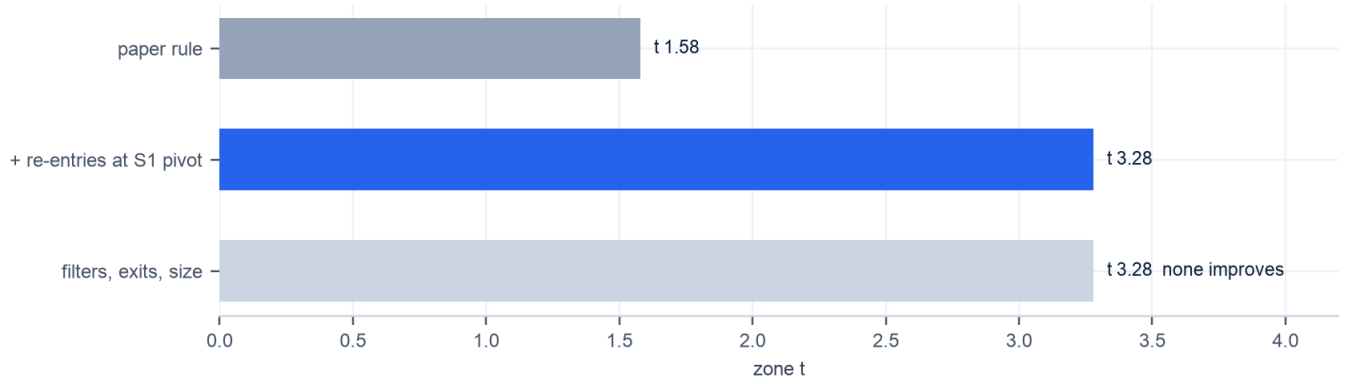


Fig. 7. Zone t at each step. The raw rule (t 1.58, drawdown $60,306) becomes a strategy only with the S1-pivot re-entries; no filter, exit or sizing rule improved it.

Before the out-of-sample period was opened, a look-ahead check compared 2,160 cuts of the data: 0 differences. The strategy was then rebuilt independently: the same 2,238 trades.

// WHAT HOLDS AND WHY IT IS NOT ADMITTED

08 ROBUSTNESS.

// STRESS TESTS ON THE WHOLE HISTORY

SCENARIO	T
as defined	3.97
double costs ($18 per trade)	3.63
entry one bar later	4.00
all times 5 minutes later	4.11
time exit 5 minutes earlier	3.88
time exit 5 minutes later	4.04

Out of sample the 5 best sessions make 16% of the net from 2024 and 46% of the net to Dec 2023.

// THE PROTOCOL

CRITERION	VALUE	OUTCOME
sample	1,502 sessions, 1,168 trades	pass
double costs	+$141,331, t 3.47	pass
robust t (HAC)	3.73	-
correction for 9.9 million tests	not significant	not passed
checks on selection (DSR, PBO, Reality Check)	not computed for the chain	not passed
base admitted	raw paper rule: exploratory	not passed

// WHY IT IS NOT ADMITTED

The base is the paper's raw rule, exploratory by construction, and once corrected for the 9.9 million tests of the batch the development result is not significant. Its out-of-sample period counts as use of the data, never as promotion.

// WHAT IT WOULD TAKE TO BELIEVE IT

09 VERDICT.

Ricochet does what most backtests do not: it stays positive on almost four years it never saw, with a stop on every trade and a simple idea behind it. Two things keep it a case study: the out-of-sample t is just below 1.96, and the rebound it was built on is weak in the data, while the re-entries do the work.

- **Forward.** Follow it on new data with the rules frozen as they are; no re-optimisation.
- **Size.** One NQ is $20 a point: on a prop account it would run on micro contracts.
- **Afternoon.** Out of sample, trades opened after noon lose (−$7,480): a known weak spot.

// IN ONE LINE

A rebound idea with a century of evidence, a stop on every trade and a positive out of sample at t 1.95: promising, not proven.

10

// THE SOURCES CITED IN THIS NOTE

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// APPENDIX · THE TERMS USED

GLOSSARY.

- **Development and out of sample.** Development (to 25 Oct 2022): the data used to choose the rules. Out of sample (from 26 Oct 2022): data never used for any choice, opened once.
- **Exploratory out of sample.** Opened for a strategy the protocol does not admit: it counts as use of the data, never as promotion.
- **t on sessions.** The mean net of the sessions with trades divided by its standard error; around 2 means the result is unlikely to be chance.
- **HAC.** A standard error robust to changing variance and correlation between nearby sessions (Newey and West, 1987).
- **AV threshold.** The minimum mean net per trade for a market, set to cover costs and slippage: $70 on NQ.
- **S1 pivot.** The first support level of the classic floor-trader pivots, computed from the previous day's high, low and close.
- **Plateau (zone t).** The t of a neighbourhood of parameters, not of the single best point.

// SIGNED

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Research note · 4 Nov 2026 · research only, not investment advice